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Advanced fire emergency management based on potential fire risk assessment with informative digital twins

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ABSTRACT

Modern large complex buildings, with their vast expanses and multipurpose usability, present diverse fire-spread scenarios, necessitating precise awareness of fire situations. Despite numerous studies integrating building information models with IoT and AI technologies to provide contextual information, there may be limitations in recognizing potential risks under dynamic fire behaviors during urgent situations. This paper presents a fire emergency management system based on an informative digital twin (IDT) to provide instantaneous fire situational information. Utilizing IDT, the system gathers building-specific fire scenario knowledge for immediate fire recognition. Leveraging this knowledge, the IDT assesses building fire risks and shares safety information on egress routes and zones with fire-related applications. A simulation case study revealed a fire recognition processing time of 0.88 s, acceptable for fire emergencies. Additionally, the IDT's risk assessment reduced the total evacuation time by 65 s and potential injuries by 406, demonstrating the system's effectiveness in fire emergency management.

1. Introduction

As modern buildings become increasingly large and complex, the risk of fire hazards increases. Large complex buildings such as shopping malls, airports, and underground shopping areas, bustling with people, are prone to significant threats to both life and property during fires [1]. In particular, these buildings, with their intricate structures and varied uses across different sub-zones [2], present a dynamic fire spread and potentially complicate the control and management responses during such emergencies [3]. Survival rate in the event of a fire relies heavily on fast on-site decision-making. However, building characteristics can obscure the assessment of the situation during a fire, increasing the difficulty for evacuees in making appropriate decisions. This can lead to delayed evacuation and serious casualties. It is challenging for building occupants to promptly evacuate based solely on simple fire alarms. Therefore, these buildings require an informative fire-response management system that can provide detailed information about the fire situation, including fire spread trends and potential risk levels. This enables the evacuees to make informed decisions quickly and formulate effective response strategies.

How do the challenges faced by evacuees in large complex buildings differ from those in more straightforward structures, and why does this complexity obstruct decision-making during a fire? The process of evacuating from a fire involves detection, recognition, response, and movement. In large complex buildings, the initial challenge for evacuees is often to observe and assess their environment rather than promptly detect and recognize a fire [4]. This tendency can be more pronounced in large complex buildings, where it is difficult to directly identify the source of the fire and where crowd psychology might lead to complacency [5]. Fortunately, the advancement of intelligent sensor technology has improved the detection accuracy [6–8], and fire monitoring systems based on building information modeling (BIM) can assist in identifying the source of a fire [9], thus minimizing delays in the early stages of fire detection and recognition.

The problem of evacuation delays in complex buildings, owing to their unique features, continues to be a challenge in the response phase following recognition. Evacuees plan escape routes before evacuation and firefighters develop strategies before conducting rescue operations. In this context, information about the spread of fire and smoke is crucial; however, because of the complex structure and various subzones of buildings, it is challenging to predict fire-spread patterns. Current fire scenes visualized in three-dimensional maps can assist; however, effective response strategies must consider the future trajectory of fire

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0926-5805/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

spread, which relies on individual experiences. There is a potential deficiency in the prerequisite knowledge required to predict the dynamic spread of fire in large complex buildings for both evacuees and firefighters [10]. This gap can lead to inefficient decisions and delayed responses, potentially causing a severe loss of life [11]. Fire simulations using computational fluid dynamics (CFD) can offer useful information [12] but are currently limited mainly to post-incident analysis and strategy development owing to constraints in execution speed [13].

The structural complexity and large scale of buildings also present challenges in decision-making during movement phases. Large complex buildings typically have multiple exits and long and intricate escape routes [14]. The occupants of large complex buildings, the majority of which are often not residents, are unfamiliar with the building layout, making it very difficult to navigate routes [15]. In the event of a fire, evacuees must continually find and move along safe paths until evacuation is complete. They cannot cross excessively hot areas, which are filled with toxic gases and heavy smoke, impairing their visibility. In other words, evacuees must identify and avoid areas with a high potential fire risk while moving. Conversely, firefighters may need to prioritize these high-risk areas to increase their rescue efficiency. However, it is not easy for both groups to assess potential risks, as this requires tracking and predicting changes in fire-related risk factors over time, such as temperature variations and carbon monoxide concentrations.

In summary, the characteristics of large complex buildings make them prone to inefficient decision-making and response strategies, potentially leading to a severe loss of life [16]. In such cases, training within a building using various fire scenarios can aid in better decision-making during fire emergencies. However, conducting realistic drills in complex buildings is often challenging because of the large number of temporary occupants and variety of factors to consider during a fire. As mentioned, real-time predictions of fire spread and information on potential hazards within a building can complement the knowledge gained from training and help derive optimal response strategies. However, this requires fire emergency management systems (FEMSs) to overcome challenges related to simulation speed and to be capable of rapidly tracking and predicting changes in fire risk factors.

As one of the technologies poised to address the aforementioned issues, a digital twin has the potential to improve information management and decision-making in various sectors, including manufacturing [17], the automotive industry [18], and construction [19,20]. It overcomes the challenges of managing dynamic, large-scale data in physical environments, which is a difficulty faced by BIM [21,22], and augments system performance by integrating sophisticated data analysis technologies, such as artificial intelligence (AI) [23] and simulation [24]. A pivotal aspect of digital twins is their utilization of virtual models and environments, enabling experiments that surpass the conventional limitations of reality, such as time, space, safety, and cost [25]. The knowledge gained from these experiments can be used to predict future changes and develop optimal response strategies [26]. Inspired by these capabilities, we devised a method to model and simulate various fire scenarios within virtual environments, gather critical insights into potential fire situations, and utilize this information to enhance decision-making in actual fire emergencies.

In this study, we proposed an informative digital twin (IDT)-based FEMS specifically designed for large complex buildings. The system employs fire-spread information and potential fire risk assessments to overcome delays in fire response due to decision-making difficulties in large complex structures.

The main contributions of this study are as follows:

(1) **IDT-based Building Fire Emergency Management:** We introduce an IDT-based FEMS architecture capable of providing fire-spread information and potential fire risk assessment in large complex buildings. This architecture leverages a cloud–edge computing framework for efficient handling of large-scale data. The IDT, built upon object-oriented design principles, functions at the edge and develops internal behavioral logic that is essential for delivering the necessary information throughout each phase of the fire response. It provides appropriate fire situation information to fire-response entities and is responsible for managing all data-processing activities.

(2) **Unique Fire Knowledge Base Construction and Utilization:** We propose a mechanism using the IDT to create a building-specific fire knowledge base, along with methods for its effective utilization. Recognizing the computational challenges of CFD, our system pre-simulates potential fire scenarios using the high-resolution building model of the IDT. During fires, the system employs a similarity comparison algorithm to compare real-time sensor data with data extracted from the fire knowledge base, thereby facilitating the immediate analysis of fire and smoke propagation trends.

(3) **Zone-based Risk Assessment Algorithm:** We propose a zone-based risk assessment algorithm based on the IDT designed to support safe fire-response plans in the event of a fire. This algorithm estimates the level of risk in different zones at specific times, considering the dynamic changes in a fire in buildings with complex structures and various utilized subzones. It uses factors such as temperature and smoke and carbon monoxide concentrations to assess risks. Evacuees can use this risk information to safely evacuate within the available safe escape time, thereby avoiding movement inefficiencies. Rescuers can use this information to determine priorities for rescue operations.

2. Literature review

2.1. Building fire management systems

In the event of a fire, to ensure the safety of occupants, the total evacuation time, known as the required safe egress time (RSET), must be shorter than the available safe egress time (ASET) [27]. This principle applies to both general buildings and large complex buildings equally. The total evacuation time comprises the detection, pre-movement, and movement times. Caution is necessary at all stages because delays at any stage can lead to overall delays. Research has been conducted on building fire management systems capable of providing appropriate fire situation information to assist individuals in making prompt decisions throughout each phase of the evacuation process.

The fire alarm system within a building provides information about a fire at the detection stage, and with the advent of Internet of Things (IoT) technology, it has become an essential piece of equipment within buildings [28]. Fire alarms are critical because they serve as primary indicators of fire presence, marking the beginning of emergency response actions. However, the effectiveness of such alarms can be compromised by frequent false alarms, which lead to public skepticism and potential evacuation delays. To address this issue, researchers [7,8] have applied AI to analyze patterns in fire sensor data to identify uncertainties and reduce the likelihood of false alarms. Another approach has enhanced the reliability of alerts by using a combination of sophisticated sensors, including smoke detectors and cameras, and leveraging an ensemble of sensory data to improve the detection accuracy [6].

Numerous studies have been conducted to provide information on fire situations, even during the post-detection stage. The authors of [29] aimed to support the development of evacuation strategies in the pre-movement stage using a BIM model to visualize fire situations collected from sensor data. In [30], the authors addressed the challenge of evacuees struggling to understand a fire environment by introducing an augmented reality (AR) device. This device facilitated awareness of the fire situation by displaying virtual arrows and providing voice guidance for evacuees along the shortest safe route to an exit. The research presented in [31] revealed that utilizing virtual reality (VR)/AR devices

to convey information about fire scenarios can significantly improve the cognitive ability of individuals to navigate dangerous situations through training. Furthermore, Kong et al. [32] and Chou et al. [33] shifted the focus toward providing essential information for firefighters, such as the locations of evacuees and the status of the fire, viewing the scenario from the perspective of rescuers rather than evacuees.

Although numerous studies have shown the potential of delivering essential fire situation information to stakeholders during fire incidents, addressing problems that may arise in large complex buildings remains a challenge. As discussed previously, the essence of ensuring safety during a fire lies in reducing the total evacuation time by minimizing delays at every stage [34]. It is not merely about the initial alarms activated by the building's detection systems; there is a pressing need to enhance fire safety features and offer appropriate information about fire situations, including visual information on fire situations, escape routes, etc. Although various research efforts have demonstrated the feasibility of supplying such information at different stages, a lack of systematic integration among technologies may result in erroneous decisions propelled by potential inaccuracies and misinformation [35]. The complex nature of large buildings requires managing a substantial volume of disaster-prevention data, which exacerbates these challenges. The capacity of technologies such as the narrowband IoT (NB-IoT) to support up to 200,000 devices is promising [36]; however, effectively processing the vast amounts of data generated during a fire to derive relevant information at each stage presents a distinct challenge.

The occurrence of complex fire patterns owing to intricate structures and various subzones complicates decision-making for evacuees and firefighters alike [15]. During a building fire, more than half the occupants reported feeling deprived of the necessary data for making informed decisions [37]. Similarly, over 83% of firefighters reported a lack of critical decision-making information. This information shortage regarding fire situations exacerbates decision-making challenges in complex fire patterns. Additionally, large buildings with complex layouts, typically featuring multiple evacuation routes, inherently experience prolonged evacuation times [38]. Ensuring the safety of any chosen route throughout the evacuation process is difficult owing to the changing conditions [14]. Under these circumstances, it is advantageous to develop an evacuation strategy that includes the prediction of fire and smoke spread trends based on comprehensive fire simulation analyses.

For instance, in [39], a fire spread model based on a fire dynamics simulator (FDS) was developed considering the complex nature of skyscrapers and the rapid spread of fires. This model used the analyzed data to address the challenges of fire prevention and determine evacuation strategies for occupants. Similarly, in the context of wildfires, where the fire trajectory is influenced by diverse materials at the ignition source, fire simulation results play a pivotal role in shaping response strategies by predicting fire spread and behavior [40]. Analyzing changes in temperature, smoke, and carbon monoxide during fire situations through simulations, as shown in a series of studies, can help identify potential hazards and develop optimal strategies. However, CFD-based fire simulations can be time consuming, often requiring hours or even days [13], which limits their immediate use during a fire event. This limitation becomes more evident as the complexity of the model increases.

2.2. Digital twins in disaster safety

A digital twin is a virtual representation that mirrors a potential or actual physical entity spanning from the micro-atomic level to the macro-geometrical level [41]. By utilizing data analysis technologies, including AI, digital twins process large datasets to enhance the monitoring, prediction, diagnosis, and optimization of physical systems [19]. Although they may be confused with BIM models, digital twins are distinct in their focus on replicating an existing physical entity and their ability to represent the real-time condition of a physical system through connectivity [21]. This real-time data integration capability supports dynamic decision-making and optimization, distinguishing digital twins from BIM models. The functionalities of data management and decision support provided by digital twins present significant opportunities for improving the efficiency of construction and asset management [42]. Furthermore, their recent deployment in disaster and safety management, leveraging these features, has attracted significant attention [20, 43].

As information structures, digital twins excel in the lifecycle management of buildings and disaster management systems by collecting and analyzing real-time data. This is particularly apparent in disaster safety management, such as that of fire accident scenarios. For instance, Khajavi et al. [44] utilized digital twins to enhance safety protocols throughout a building's lifecycle by focusing on fire detection and anomaly identification. Wang et al. [45] emphasized that the effectiveness of a digital twin lies in its capacity to faithfully represent data and their meanings, and to harness the entirety of these data for knowledge processing. A recent study [46] investigated the integration of digital twins and semantic web technologies for the intelligent management of building fire prevention systems, developing digital twins that combine static building geometric data with dynamic sensor data for proactive fire prevention management.

By facilitating information processing, digital twins can be integrated with advanced analytical tools, such as AI and simulation, to provide stakeholders with the analyzed information they need. Zhang et al. [47] proposed an AI-based digital fire framework to develop a digital twin that can provide detailed information such as temperature profiles within buildings, fire locations, and fire sizes. This system, named AID-Fire, is expected to be utilized for fire-trend prediction and evacuation guidance based on AI technology, demonstrating the potential to provide analyzed information within buildings for fire response. Ding et al. [23] identified a lack of real-time dynamic evacuation information, such as the location of occupants and the status of firefighters during the evacuation and rescue stages, as one of the main causes of casualties in building fires. Their study explored smart human monitoring technology based on digital twins using AI, which employed a Convolutional Neural Network(CNN) -based evacuee detector and multi-object tracking algorithm to assist firefighters in accurately locating evacuees.

Simulation-based prediction and analysis are the core aspects of digital twin technology [48,49]. Leveraging this capability, research has been conducted to offer predictive insights into disaster safety domains, including smart cities and wildfires. Ford et al. [50] proposed a digital-twin-based smart disaster management system known as SCDT by integrating simulation technology across various smart city infrastructures. The authors highlighted that the SCDT could virtually navigate the disaster management cycle using simulations, enhance predictions of expected action outcomes, and facilitate feedback on the ramifications of the actions implemented. Yun et al. [25] developed a novel digital twin framework designed to enhance the precision of disaster monitoring and prediction and mitigate the impact of natural disasters. In this framework, they introduced a hybrid disaster model that combined a physics-based model with a data-driven error correction model to improve disaster prediction accuracy.

Although several studies have been conducted using digital twins, research on their use to solve fire-response scenarios is less common. Moreover, many of these studies do not provide a comprehensive view of the entire process but rather narrow the focus to specific response steps [7]. Such a fragmented approach can make it difficult to adequately provide the required information at different stages of a fire emergency. In addition, as has been repeatedly emphasized, FEMSs rely on intelligent tools for prediction and analysis, and to be effectively utilized in emergency fire scenarios, they must overcome limitations in processing speed.

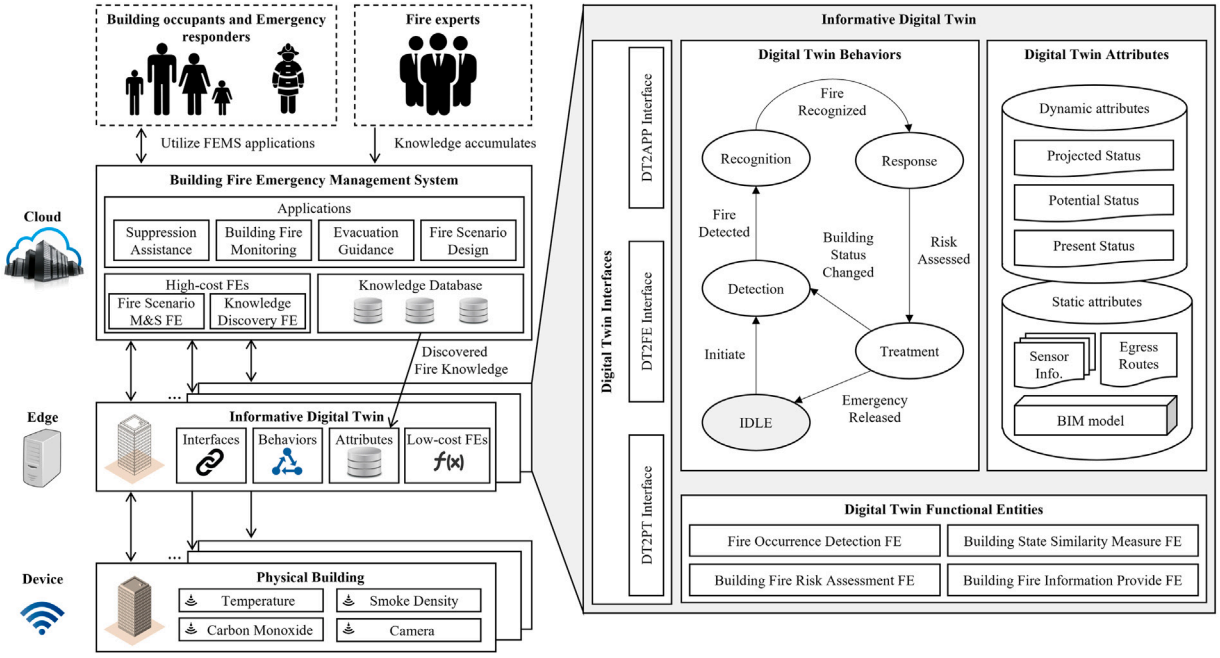


Fig. 1. Overall architecture of the IDT-based building fire emergency management system.

2.3. Goals of the study

This study aims to explore the design and implementation of an FEMS that leverages an IDT to provide appropriate fire situation information during the fire-response stage. The main research questions (RQs) are as follows:

RQ1 How can we develop an FEMS that provides fire situation information to support rapid decision-making by fire stakeholders throughout the steps of a fire emergency in large complex buildings?

RQ2 What are the possible approaches to obtaining immediate information on fire and smoke spread trends during ongoing fire situations rather than post-event?

RQ3 How can potential risk information from dynamic fire patterns be derived to support the establishment of safe evacuation and rescue strategies for large complex buildings?

3. IDT-based building fire emergency management

3.1. Overall architecture

The objective of the IDT-based FEMS is to assist stakeholders in making timely and effective decisions during fire emergencies. Fig. 1 shows the architecture of the system. This system employs a strategy of knowledge accumulation and discovery to overcome the limitations associated with the processing speed of CFD simulations. Prior to a fire incident, fire safety experts design and simulate various potential fire scenarios tailored to the building. In the event of an actual fire, the system leverages the IDT to search among these pre-simulated scenarios for those that closely align with the ongoing fire conditions. These selected scenarios are then used to track and predict dynamic changes in the fire situation, enabling stakeholders to rapidly make informed decisions.

The IDT is a comprehensive digital representation of a physical building that plays a pivotal role in building fire emergency management. The IDT-based system is designed to utilize cloud–edge computing, enabling the efficient processing of vast amounts of data generated by various fire safety equipment in buildings.

(1) Cloud-computing layer. The cloud-computing layer offers computing resources for processing high-cost functional entities (FEs), handling vast fire knowledge bases, and executing building fire emergency management applications. Equipped with high-performance computing and ample storage space, cloud-computing servers enable fire experts to design and simulate numerous potential building fire scenarios with high resolution. To accommodate numerous users, FEMS applications are hosted on the cloud for efficient access.

(2) Edge-computing layer. The IDT operates within the edge-computing layer and processes raw sensor data to derive appropriate fire situation information for applications. By preprocessing the data streams generated by numerous sensors in a building at the edge, the IDT eliminates the need to send unnecessary data to the cloud, thereby avoiding bandwidth waste that could occur as the number of buildings managed by the system increases [51,52]. The IDT follows predefined internal response logic steps to derive the fire situation information required by the applications. Because the IDT is responsible for executing the necessary processes at each stage of a fire emergency, tracking the progress and timing of the fire response is straightforward.

(3) Device layer. The device layer comprises a multimodal sensing network powered by various IoT sensing devices spread throughout a building. These devices collect critical sensor data regarding the physical building environment, including temperature, smoke density, and carbon monoxide levels. These sensor data are relayed to the edge-computing layer and used to analyze the building fire situation.

3.2. Informative digital twin

The IDT was designed to adhere to object-oriented design principles. The principles of modularity and encapsulation enhance the flexibility and scalability of the FEMS. On this basis, four modules (i.e. attributes, behaviors, FEs, and interfaces) were established, as illustrated on the right side of Fig. 1. The following sections provide a detailed description of each module.

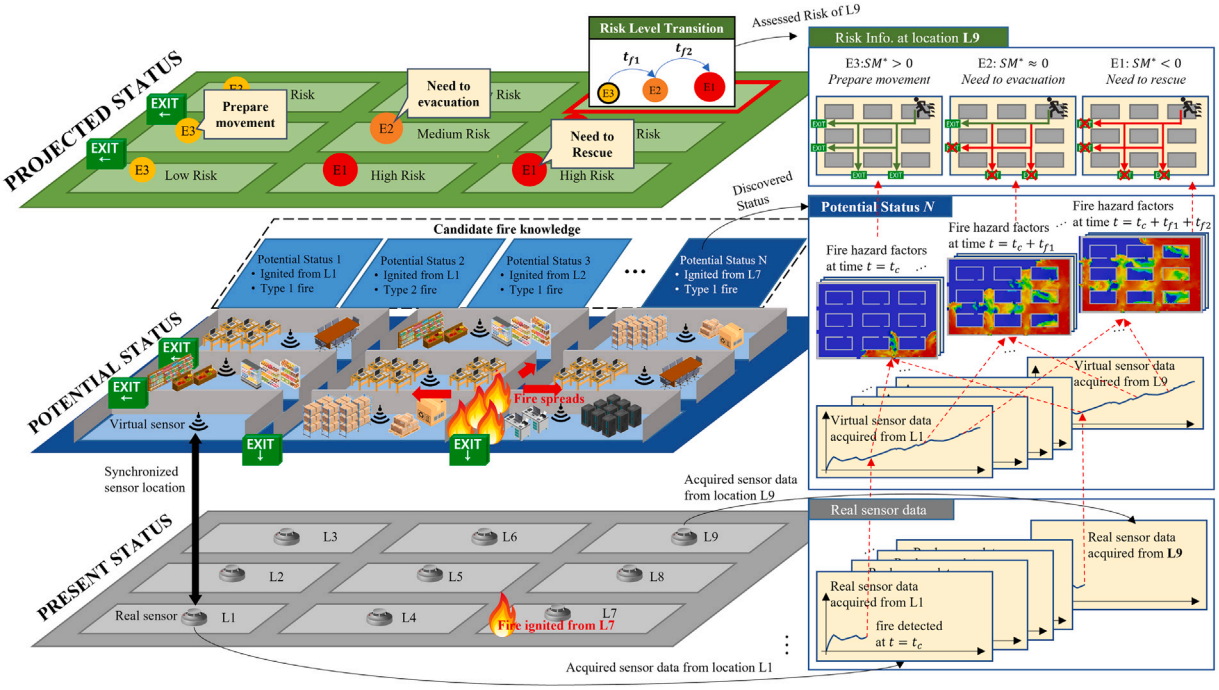


Fig. 2. Dynamic attributes of an informative digital twin.

3.2.1. IDT attributes

IDT attributes are a collection of information that represents the physical building status from a multidimensional perspective. These attributes are categorized as static or dynamic. Static attributes, which include the BIM model and sensor metadata (e.g. location and type), remain constant unless they are manually modified. These static attributes are used to design and simulate potential fire scenarios. In contrast, dynamic attributes adjust over time in response to changes in a building's status and primarily emerge during a fire emergency. Dynamic attributes are categorized into the following three parts:

- Present status. The present status represents the building's current status, which includes the temperature, smoke, and carbon monoxide data observed by IoT sensor devices.
- Potential status. The potential status denotes the possible status of the building from the present to the future. It includes virtual sensor data synchronized with real sensors and fire hazard factors representing the expansion and migration of fire products such as gas temperature, smoke visibility, and carbon monoxide concentration.
- Projected status. The projected status represents the potential risk information derived from the potential status matched to a real fire scenario. For each egress route and zone, the projected status includes the available time, required time, and safety margin for safe evacuation and rescue.

According to the operation of IDT behaviors during a fire emergency, the present, potential, and projected statuses are linked together, as shown in 2. Assuming that a fire ignited from location L7, the present status attribute holds the real sensor data from the time of fire detection to the current point in time t_c. The potential status attribute contains numerous potential statuses discovered from the fire knowledge base, each of which contains virtual sensor data from the time the virtual fire was ignited to the end of the virtual fire scenario. Because each virtual sensor has a position and type synchronized with a real sensor, it is possible to discover potential statuses that match reality by comparing the sensor data generated by each sensor. The potential status includes fire hazard factors that vary with fire dynamics and are distributed throughout the internal area of the building. Fire hazard factors are represented as spatiotemporal data, including gas temperature, CO concentration, and smoke visibility, and are utilized to assess the fire risk in each area. From the discovered potential status, the projected status represents the risk level at the current time and the remaining time to the next risk level. As the aligned potential status includes fire hazard factors in the future, the remaining time for the next risk level can be derived. For example, at the current time t_c, the egress routes from location L9 to all exits are tenable, and there is sufficient time to prepare for movement and escape. At a point in the future after time t_{f1}, only one escape route is tenable; hence, it must be used to escape as quickly as possible within a limited time. Therefore, the preparation movement action must be performed within t_{f1}. Similarly, after $t_{f1}+t_{f2}$, no egress routes are available; therefore, the escape action must begin within t_{f2} at time $t_c + t_{f1}$.

3.2.2. IDT behaviors and functional entities

IDT behaviors comprise a set of processes designed to support fire emergency management by automatically responding to dynamic changes in the status of a physical building. These behaviors are organized into a state machine with five states: idle, detection, recognition, response, and treatment. As the status of the physical building changes, the IDT seamlessly transitions its behavioral states and activates the FEs associated with the current state. FEs are computing components that are invoked according to behavioral logic and process data according to each function and purpose by interacting with the fire knowledge base and IDT attributes. These FEs are strategically deployed to ensure smooth operation of IDT behaviors. Considering the computing and storage space requirements of each FE, some FEs with high computational costs are integrated into the cloud-computing layer, and other FEs that have low computational costs and are forced in real time are integrated into the IDT in the edge-computing layer.

Fig. 3 shows the detailed operating mechanism of the IDT-based FEMS.

(1) **Idle:** Accumulate fire knowledge to recognize actual fire situations (implementation details are provided in Section 4.3.1)
- When generating knowledge (1-1), experts use static attributes (1-2) to model and simulate various fire scenarios and accumulate them as knowledge (1-3).

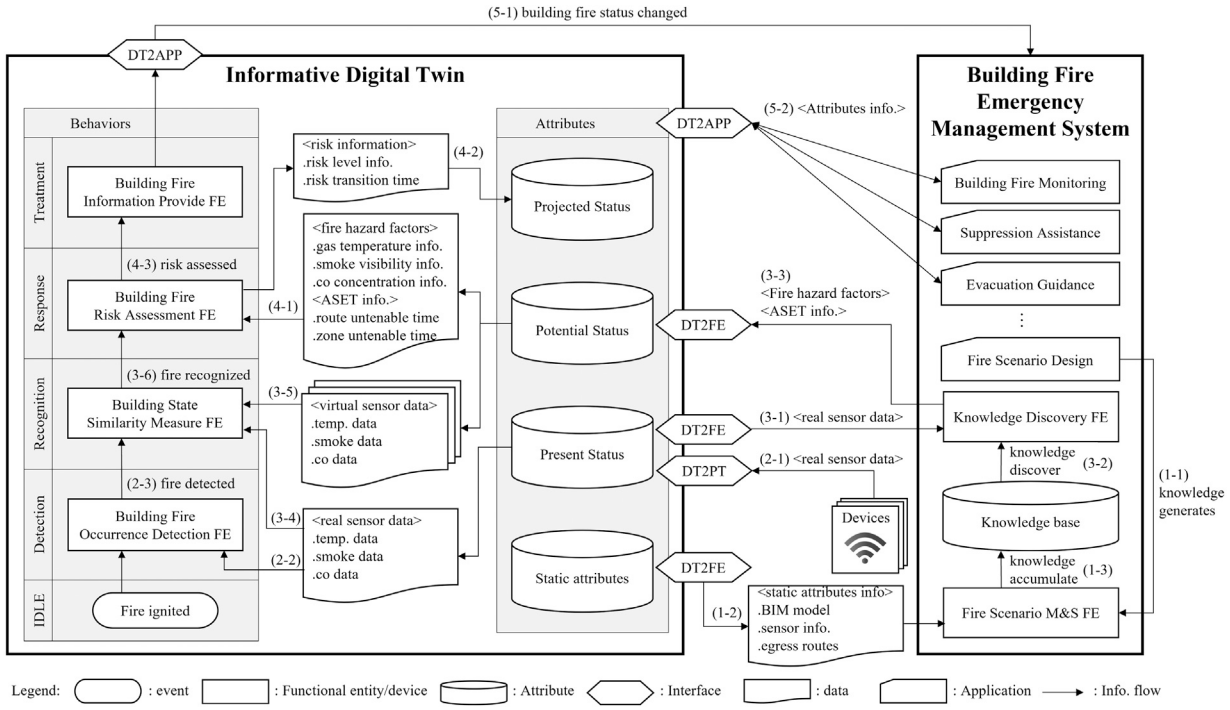


Fig. 3. Operating mechanism of the IDT-based building fire emergency management.

(2) **Detection:** Detecting whether a fire has occurred
- Collect real sensor data (2-1), detect a fire (2-2), and notify the next status when a fire is detected (2-3).

(3) **Recognition:** Recognize the progress of a building fire (implementation details are provided in Section 4.3.2)
- Discover knowledge from the fire knowledge base (3-2) based on real sensor data (3-1) and store potential fire situations (3-3). Derive the most similar scenario (3-6) through a similarity measure between real sensor data (3-4) and virtual sensor data (3-5).

(4) **Response:** Risk level assessment of recognized situations (implementation details are provided in Section 4.3.3)
- Receive the fire hazard factor for the fire scenario derived with the highest similarity (4-1), assess the fire risk and save (4-2), and notify the next status (4-3).

(5) **Treatment:** Treatment for fire stakeholders (FEMS applications are described in Appendix A)
- Notify applications that information is available (5-1), and applications access attributes to utilize fire situation information (5-2)

3.2.3. IDT interfaces

IDT interfaces are communication modules that facilitate communication between an IDT and its external components. Depending on the connection target, IDT interfaces are categorized as digital twin to physical things (DT2PT), digital twin to functional entities (DT2FE), and digital twin to application (DT2APP) interfaces. The DT2PT interface supports connectivity between the IDT and the physical buildings. It receives measured data from numerous IoT sensors distributed in large complex buildings and stores the obtained sensor data in the IDT's present status attribute. The DT2FE interface supports information exchange between the IDT and the high-cost FEs of the cloud-based building FEMS. Through the DT2FE interface, the IDT invokes the knowledge discovery FEs and receives relevant fire knowledge, and the fire scenario modeling and simulation (M&S) FE accesses the static attributes of the IDT. The DT2APP interface supports building fire emergency management applications in accessing the IDT attribute information. Through the DT2APP interface, various building fire emergency management applications can access the IDT attributes and receive appropriate information according to their purpose. A detailed implementation of the IDT interfaces is described in Sections 4.1 and 4.2.

4. System development

4.1. Development Kubernetes/KubeEdge-based FEMS system

To implement the edge-cloud system architecture presented in Section 3.1, we utilized Kubernetes [53] and KubeEdge [54] to construct a cloud–edge computing environment, as illustrated in Fig. 4. In our environment, the cloud consisted of two servers, and the edge consisted of one x86 PC. The IDT process, along with the low-cost FEs, runs at the edge, whereas fire management applications and high-cost FEs operate in the cloud. Parallel processing can be optionally performed for faster speed enhancement in simulations and knowledge discovery when necessary. The Kubernetes controller can automatically place containers based on the resource demand and availability, thus making the parallel processing configuration easier. The knowledge DB is managed using a persistent volume subsystem, and the processes in the cluster can be accessed by the DB.

Considering the real-time communication between the IDT and high-cost FEs, processes running inside containers are interconnected based on the Data Distribution Service (DDS) [55]. The deliverability of data can be efficiently managed by considering quality of service (QoS) factors, such as deadlines. IoT sensor data are collected through an edge broker server based on the Message Queuing Telemetry Transport (MQTT) protocol [56], and the processed IDT attributes are provided to the FEMS applications through a REST API server.

4.2. Development of object-oriented informative digital twin

A UML class diagram of the developed IDT is shown in Fig. 5(a). Based on the object-oriented design principles and IDT design principles described in Section 3.2, the key components of the IDT are well established. These components include the IDT object itself, four attributes,

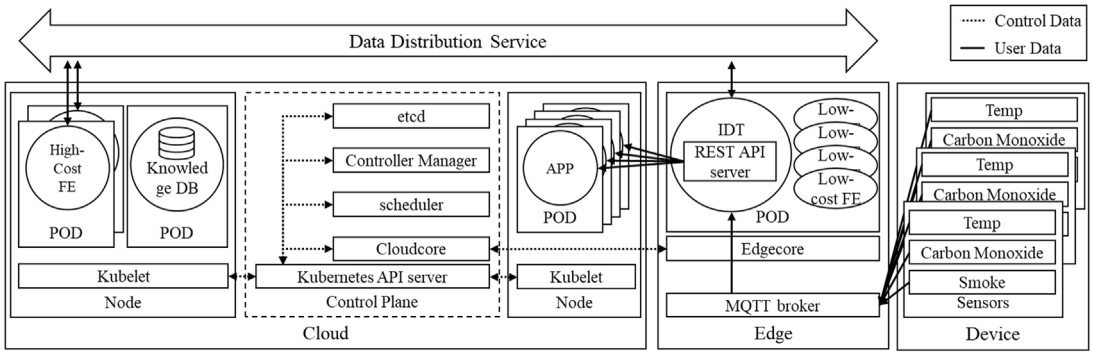
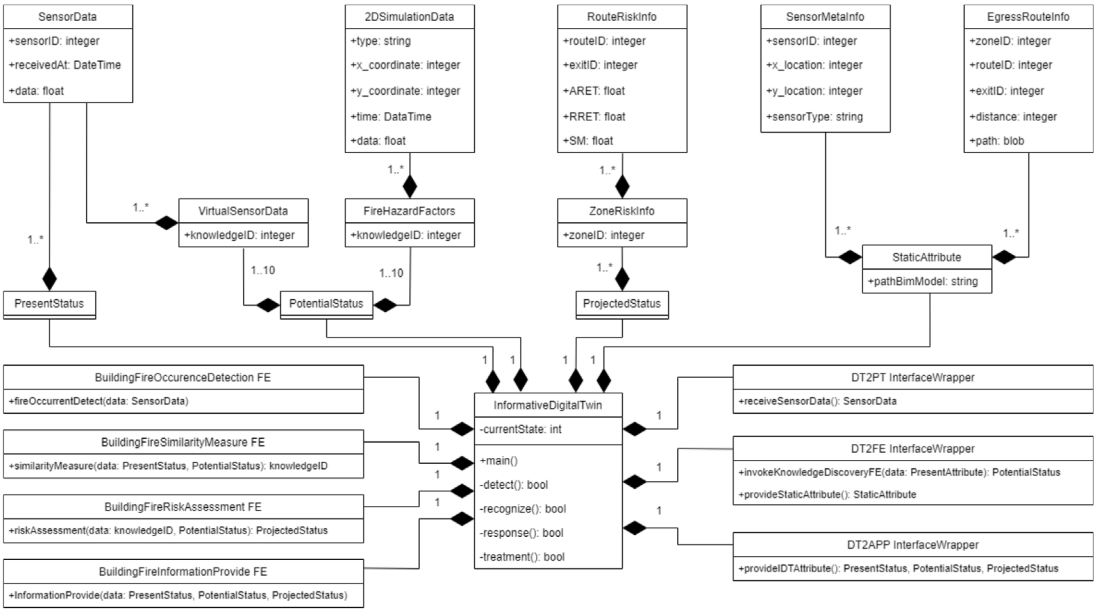
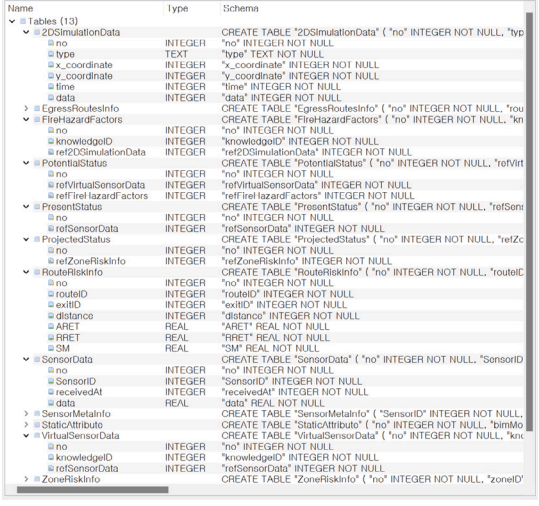


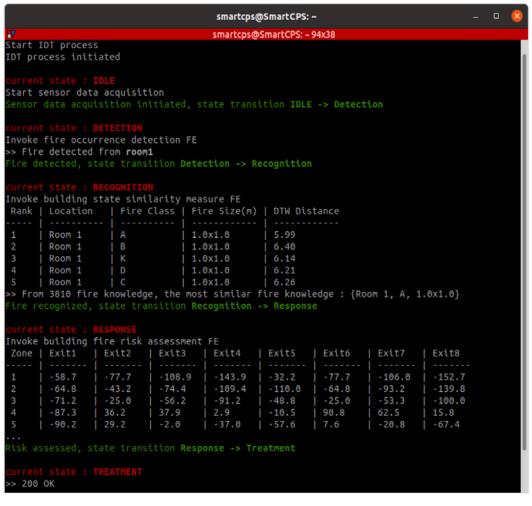
Fig. 4. Implementation of FEMS system based on Kubernetes/KubeEdge.



(a) Class diagram.



(b) IDT attributes.



(c) IDT behaviors.

Fig. 5. Class diagram, attributes, and behaviors of the developed informative digital twin.

Table 1
Request examples of DT2APP interface based on REST API. The prefix * is http://<ipaddress>:<port>.

HTTP request	Description
`GET*/PresentAttribute/SensorData`	Obtain all real sensor data acquired by the IDT
`GET*/PotentialAttributes/FireHazardFactors?knowledgeID=1`	Obtain all fire hazard factors of fire knowledge with ID = 1
`GET*/ProjectedAttributes?zoneID=1&exitID=12`	Obtain all risk information corresponding to Zone 1 and Exit 12 in projected attributes

four FEs corresponding to the behaviors, and three interfaces. Each component interacts with the IDT behavior according to its unique role. In the class diagram, the attribute components describe the data structure managed by the IDT and the FE components process and return the received data according to their given roles. The IDT interfaces serve as communication interfaces with the IDT external components and are implemented through the Wrapper Class. The relationships between the components are indicated by solid arrows in Fig. 5(a).

To store and manage the IDT attributes data, we used SQLite [57], which is a lightweight and server-less relational database engine that supports efficient data management in resource-constrained edge environments. The data structure of the IDT attributes is constructed based on the definitions of dynamic (present, potential, and projected statuses) and static attributes. The present status continuously stores real-time data observed from various sensors distinguished by the sensor ID, along with timestamps (receivedAt). The potential status holds fire knowledge composed of virtual sensor data and fire hazard factors, each identified by a knowledge ID. Sensor data for virtual sensor data are designed to match the sensor data structure of the present status for seamless similarity measurements. Fire hazard factors, represented in two dimensions, consist of multiple two-dimensional simulation data entries stored over time for each (x, y) coordinate. The projected status includes zone risk information, which is distinguished from the zone ID. Each zone risk information contains multiple route risk information entries, which store available route egress time (ARET), required route egress time (RRET), and safety margin (SM) data for escape routes between specific exits (exit ID) and within the zone (zone ID), whereas the static attribute manages sensor meta information, which includes the location (x, y) and type of one or more sensors for designing potential fire scenarios. It also manages information on one or more egress routes, representing the trajectories of escape routes and distances, as well as the file path corresponding to the BIM model. Fig. 5(b) shows the developed IDT attribute based on SQLite.

The IDT behavior was modeled using the Business Process Model and Notation (BPMN) [58], which is an Object Management Group (OMG) standard for specifying processes in a workflow. The IDT behavior contains four states, namely, detection, recognition, response, and treatment, and invokes each FE component according to its current state, as shown in Fig. 5(c). The detailed operation of the FE components is presented in Section 4.3.

To implement the DT2PT, DT2FE, and DT2APP interfaces, we utilized the MQTT, DDS, and REST API, respectively. In DT2PT interface, each real sensor data is structured as `{datetime, data}` and collected through the MQTT topic of `sensordata/<sensor_id>`. In the DT2FE interface, the static attributes required to generate potential fire scenarios on the cloud side are transmitted through the DDS topic in the `static_attribute/`, and requests and responses for discovering similar fire knowledge are transmitted via the `discovered_knowledge/` DDS topic. For the DT2APP interface, REST APIs are designed to access the IDT attributes, as shown in Table 1.

4.3. Development of core FEs and fire knowledge base

4.3.1. Building fire scenario modeling and simulation

The goal of building fire scenario modeling and FE simulation is to construct an extensive fire knowledge base for fire recognition. This is achieved by simulating a range of potential fire scenarios within a building, considering its physical attributes and different fire conditions. By accumulating various potential fire scenarios in the fire knowledge base, IDT systems have more opportunities to identify potential fire scenarios that more closely reflect reality, thereby improving the fire recognition accuracy.

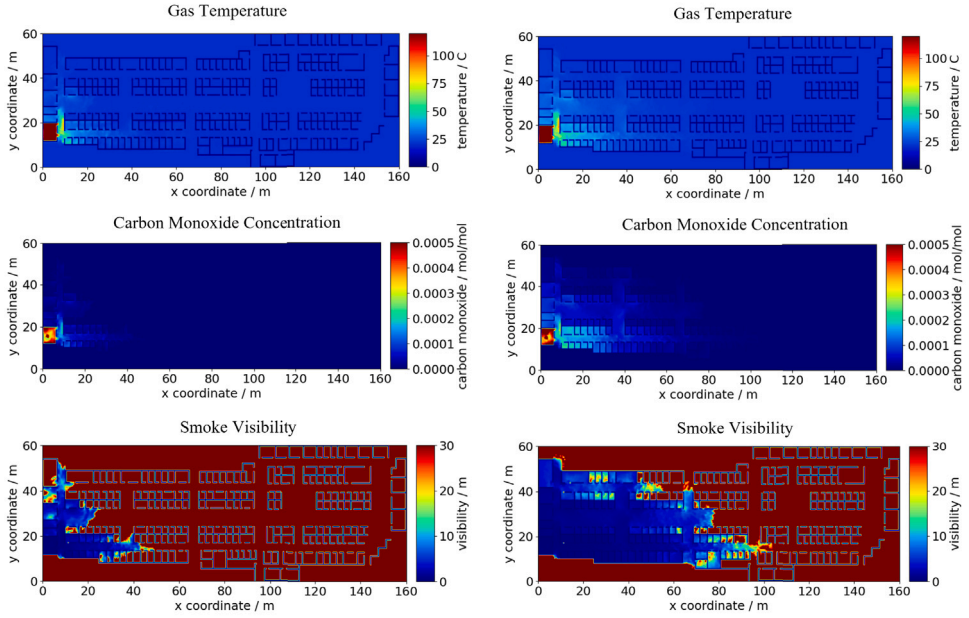
To simulate building fire dynamics, we utilized an FDS, which was developed by the National Institute of Standards and Technology (NIST) and is primarily used in the fire safety domain. In the modeling phase, the BIM model and sensor information in the static attributes of the IDT are used to reflect the physical properties of the building. The development of Industry Foundation Classes (IFC), which is an open data model standard (ISO 16739), facilitates the interoperability between BIM and the FDS. Building fire scenario modeling is performed using the following steps:

1. The BIM model is imported into BlenderFDS, a third-party FDS modeling tool for converting IFC to an FDS. As a result of the conversion, the building geometry is represented by OBST symbols in the FDS.
2. The location and properties of the virtual sensor devices are defined based on actual sensor information. In the FDS model, virtual sensor devices are represented as DEVC symbols with locations specified by XYZ coordinates and properties specified by QUANTITY parameter.
3. The thermal properties of the building material are defined based on the ASTM E84 Fire Standards [59]. The thermal properties are represented as MATL symbols and applied to the surface (SURF) of the building geometry (OBST).
4. The combustion conditions of a virtual building fire are defined by specifying the fire location and reaction. The location of the fire is specified by the VENT symbol, and the reaction of the fire is represented by the REAC symbol with the type of fuel.

After the modeling process is completed, slice script blocks (SLCF) are added to the FDS model to analyze the two-dimensional spatial distribution of the gas temperature, carbon monoxide concentration, and smoke visibility.

In the simulation phase, the FDS software is run using the FDS model developed above. According to script blocks DEVC and SLCF, the FDS software output virtual sensor data and fire hazard factors, respectively. The virtual sensor data are represented as one-dimensional time-series data corresponding to each IoT sensing device, which has measurements of temperature, smoke density, and carbon monoxide. Conversely, the fire hazard factors are expressed as two-dimensional spatiotemporal data, where each axis represents the x-coordinate, y-coordinate, and time. Fig. 6 shows an example of fire hazard factors (gas temperature, carbon monoxide concentration, and smoke visibility) at time points (a) 60 s, and (b) 300 s. The fire hazard factors are then used to assess the fire risk level and transition time by determining the ARET and RRET.

The fire knowledge base was constructed based on TimescaleDB, a time-series database, to efficiently store and manage the derived time-varying virtual sensor data and fire hazard factors. TimeScaleDB, which is specifically optimized for time-series data, offers efficient data ingestion and querying performance, which are essential for dealing with large volumes of continuously evolving data. Its automatic data partitioning and time-based query optimization significantly enhance the query performance and data management. Moreover, TimeScaleDB's ability to scale effortlessly and its advanced data compression and retention policies make it ideally suited for building an extensive fire knowledge base, ensuring the efficient storage and rapid retrieval of



(a) at 60 s. (b) at 300 s.

Fig. 6. Fire hazard factors (a) at 60 s and (b) at 300 s.

Table 2
1D virtual sensor data.

Column name	Data type	Properties
knowledge_id	INT	NOT NULL
time	TIMESTAMPTZ	NOT NULL
sensor_id	INT	NOT NULL
data	DOUBLE	NOT NULL

Table 3
2D fire hazard factors.

Column name	Data type	Properties
knowledge_id	INT	NOT NULL
time	TIMESTAMPTZ	NOT NULL
x_coordinate	INT	NOT NULL
y_coordinate	INT	NOT NULL
gas_temperature	DOUBLE	NOT NULL
smoke_visibility	DOUBLE	NOT NULL
co_concentration	DOUBLE	NOT NULL

fire knowledge over extended periods. The virtual sensor data and fire hazard factors are stored and retrieved using the standard Structured Query Language (SQL) supported by TimescaleDB.

Tables 2 and 3 list the structures of the 1D virtual sensor data and the 2D fire hazard factors, respectively.

4.3.2. Building fire status similarity measurement

The fire knowledge discovery process is performed as follows. First, by invoking the knowledge discovery FE, a fixed amount of candidate fire knowledge with high similarity is discovered from the fire knowledge base and imported to the potential status attributes. Subsequently, the building fire status similarity measure FE identifies the fire knowledge with the highest similarity to the potential status attributes. In this process, the potential status attributes act as a type of cache memory, and the similarity measure FE can be performed with a low computational overhead owing to the fixed-size search space.

To implement the two FEs above, we considered the dynamic time warping (DTW) algorithm, which is a nonlinear mapping technique used to measure the similarity between two temporal sequences. Unlike traditional similarity measures based on the Euclidean distance, DTW first allows nonlinear mapping to obtain an optimal alignment between two sequences, thereby enabling accurate similarity measurements that minimize errors due to the presence of time distortions and time shifts [60]. Let the real sensor data $RS = (X_1, X_2, \ldots, X_N)$ and virtual sensor data $VS = (X'_1, X'_2, \ldots, X'_M)$ have different lengths N and M, respectively. DTW constructs a cost matrix D of size $N \times M$ where $D(i, j)$ is the cost of aligning the ith element of RS with the jth element of VS. The cost matrix D is computed recursively using the following formula:

$$D(i,j) = d(X_i, X'_j) + \min \begin{cases} D(i, j-1), \\ D(i-1, j-1), \\ D(i-1, j) \end{cases} \quad (1)$$

$$D(0,0) = 0, D(i,0) = D(0,i) = \infty \quad (2)$$

where the distance function $d()$ is any distance metric. In this study, $d(X_i, X'_j)$ is defined as

$$d(X_i, X'_j) = \sqrt{(X_i - X'_j)^2} \quad (3)$$

and consequently, the DTW distance can be calculated as

$$DTW(RS, VS) = D(N, M) \quad (4)$$

DTW enables accurate similarity measurements by calculating the cost matrix; however, there are some issues with applying the original DTW to the knowledge discovery process. During a fire emergency, real sensor data are continuously collected to reflect the status of the building. This means that the knowledge discovery FE and the building status similarity measure FE are invoked repeatedly whenever new observation data are added. Because the original DTW outputs the similarity distance only when a cost matrix of size $N \times M$ is completely calculated, the time complexity is $O(kNM)$ at every time tick t when k fire knowledge exists. Therefore, the computational cost increases significantly as new observation data accumulate.

To address these technical issues, we modified the original DTW into an online version. As presented in Algorithm 1, online DTW calculates the cost matrix in a cumulative manner using the following steps:

Algorithm 1 Online Dynamic Time Warping

Require:
X_t be the observation data at time t
$VS =< X'_1, X'_2, ..., X'_M >$
Let:
$d()$ be the distance between two points
$D[N][M]$ be the cumulative cost matrix
function ONLINE DTW(X_t, VS)
▷ Initializing the first column for cost matrix D
if $t == 1$ **then**
$D[0][0:M] \leftarrow 0$
end if
▷ Initializing the new column for cost matrix D
$D[t][0:M] \leftarrow \inf$
▷ calculate cost matrix D for new column
for $i = 1$ to M **do**

$$D[t][i] = d(X_t, VS[i]) + \min \begin{cases} D[t-1][i], \\ D[t][i-1], \\ D[t-1][i-1] \end{cases}$$

end for
return D[t][M]
end function

(1) From the detection of a fire such that $t = 1$, a cost matrix with the first column in which the first element is zero and the remaining elements are infinite is constructed.
(2) A new column associated with time t is initialized to infinity and concatenated into cost matrix D.
(3) Each element of the new column is then calculated according to Eq. (1).
(4) Return similarity distance with $D[t][M]$.
(5) When a new observation X_t is added to the RS, online DTW is invoked again, and steps (2)–(4) are repeated.

The time complexity for invoking the online DTW is only $O(kM)$. Because the similarity distance can be obtained from the cost matrix accumulated at each time tick, the online DTW is not affected by the length of the accumulated observation data.

4.3.3. Building fire risk assessment

In this study, the ARET and RRET were introduced to assess the fire risk for each route and zone. The ARET signifies the time allocated to each evacuation route to ensure safe evacuation. Specifically, it is the duration from the moment a fire is ignited until the egress route becomes untenable to occupants. The ARET is defined as

$$ARET_{z,p} = \min(T_{CO}, T_{temp}, T_{vis}) \tag{5}$$

where $ARET_{z,p}$ represents the time at which the egress route p between Zone z and a certain exit becomes untenable for the occupants. T_{CO}, T_{temp}, and T_{vis} represent the time from fire ignition to when the carbon monoxide, gas temperature, and smoke visibility on route p reach the untenability criteria, respectively. The untenability criteria may differ according to building characteristics. NFPA 101 [61], the SFPE Handbook of Fire Protection Engineering [62], and certain studies [63] provide guidelines for defining untenability criteria. Table 4 provides the untenability criteria considered for calculating the ARET in this study.

While the ARET determines when a specific route becomes untenable, the RRET determines the time required to travel through that route at a specific point in time. The RRET can be calculated by considering the occupant movement speeds and route distances, as shown in Eq. (6)

$$RRET_{z,p}(t) = \sum_{i \in p} \frac{d_i}{v_i(t)} \tag{6}$$

Table 4
Untenability criteria.

Life safety parameter	Tenability acceptance condition
Smoke layer height	2.0 m
Smoke visibility	≥10 m of visibility
Air temperature	≤100 °C
Carbon monoxide concentration	≤2800 ppm

where $RRET_{z,p}(t)$ represents the time required to travel through egress route p between Zone z and a specific exit at a specific point in time t. d_i is the distance of the ith trajectory on route p, and $v_i(t)$ is the movement speed required to travel along the ith trajectory at time t.

During a fire emergency, hazardous factors such as air temperature, smoke obscuration, and toxic gases impair occupants' physical movement abilities and psychological conditions. Furthermore, extended exposure to such hazardous factors can significantly delay or even impede evacuation. For movement speed, we followed the approach proposed in [64], which describes the occupant moving speed as a function of three major fire hazard factors (air temperature, smoke visibility, and concentration of carbon monoxide). Eqs. (7)–(10) describe the calculation of movement speed $v_i(t)$.

Specifically, from the reference speed, which representing the occupant movement speed under unimpeded conditions, the movement speeds $v_i(t)$ on the ith trajectory are affected by the impact functions $f_i(t)$, which are influenced by the fire hazard factors of smoke visibility $vis_i(t)(m)$, the concentration of carbon monoxide $CO_i(t)(\%)$, and air temperature $Temp_i(t)(°C)$. t_{exp} represents the exposure time (min) in the toxic gas-covered area. v_0 and v_{max} are the reference movement speed (i.e. representing the evacuee's moving speed under unimpeded conditions) and the maximum movement speed of the occupants, respectively, which are commonly set to 1.2 m/s and 4 m/s, respectively [65].

$$v_i(t) = v_0 \times f_i^{vis}(t) \times f_i^{CO}(t) \times f_i^{Temp}(t) \tag{7}$$

$$f_i^{vis}(t) = \begin{cases} 1 & vis_i(t) \geq 7.5 \text{ m} \\ 1.375 - \frac{2.8125}{vis_i(t)} & 2.5 \text{ m} \leq vis_i(t) < 7.5 \text{ m} \\ 0.25 & vis_i(t) < 2.5 \text{ m} \end{cases} \tag{8}$$

$$f_i^{CO}(t) = \begin{cases} 1 & CO_i(t) < 0.1 \\ 1 - (0.2115 + 1.788CO_i(t)) & \\ \quad \times CO_i(t) \times t_{exposure} & 0.1 \leq CO_i(t) < 0.25 \\ 0 & CO_i(t) \geq 2.5 \end{cases} \tag{9}$$

$$f_i^{Temp}(t) = \begin{cases} 1 & Temp_i(t) \leq 30 \text{ °C} \\ 1 + \frac{(v_{max} - v_0) \times (\frac{Temp_i(t) - 30}{60 - 30})^2}{v_0} & 30 \text{ °C} < Temp_i(t) \leq 60 \text{ °C} \\ \frac{v_{max}}{v_0} \left[1 - (\frac{Temp_i(t) - 60}{120 - 60})^2\right] & 60 \text{ °C} < Temp_i(t) < 120 \text{ °C} \end{cases} \tag{10}$$

After calculating both the ARET and RRET, it is possible to evaluate the safety margin of route p by subtracting the RRET and the elapsed time t from the ARET as follows:

$$SM_{z,p}(t) = ARET_{z,p} - (t + RRET_{z,p}(t)) \tag{11}$$

Finally, based on the safety margin $SM_{z,p}(t)$, we can assess the risk information for Zone z by determining the safest route and the safety margin of that route.

$$p_z^* = \arg\max_{p \in P'_z} SM_{z,p}(t) \tag{12}$$

$$SM_{z,p}(t)^* = \max_{p \in P_z} SM_{z,p}(t) \tag{13}$$

Table 5
Consideration of fire class, fire size, and fire-initiation location to construct the fire knowledge base.

Fire class	Fire size (m)	Fire locations
[A, B, C, D, K]	[0.5 × 0.5, 1.0 × 1.0, 2.0 × 2.0]	Inside each room (216 points) Uniformly distributed in corridors (38 points)

where p_z^* is the safest route, $SM_{z,p}^*$ is the safety margin of the safest route, and $\mathcal{P}_z$ represents the egress routes between Zone z and the escape exits.

The safety margin of the safest route p_z^* in Zone z provides the following information to determine the safety of Zone z:

1. If $SM_{z,p}(t)^* > 0$, the risk level in Zone z is considered to be low, indicating that there is sufficient time for evacuation, and prepare for movement action must be made within the time frame of $|SM_{z,p}(t)^*|$.
2. If $SM_{z,p}(t)^* \approx 0$, the risk level in Zone z is determined to be medium, meaning that all egress routes between Zone z and the escape exits will be blocked immediately. Consequently, the occupants located in Zone z must begin evacuating.
3. If $SM_{z,p}(t)^* < 0$, the risk level in Zone z is assessed as high, indicating that all egress routes between Zone z and the escape exits have already been blocked, and time $|SM_{z,p}(t)^*|$ has elapsed after blocking. Therefore, occupants located in Zone z must be rescued with priority based on $|SM_{z,p}(t)^*|$.

5. Case study

To determine whether the proposed IDT-based FEMS can support fire responses in large-scale complex buildings (to answer RQ1), we conducted a case study focusing on an underground shopping mall. This case study also examined the effectiveness of fire knowledge discovery algorithms (to answer RQ2) and evaluated the performance of the zone-based risk assessment algorithm (to answer RQ3).

5.1. Case building description

The BIM model of the large complex building adopted in this case study was obtained from the Hoehyeon Underground Shopping Center in Jung-gu, Seoul, Korea. As shown in Fig. 7(a), the total building area is approximately 9031 m^2, with an average height of 2.4 m, width of 166 m, and length of 58 m. A total of 987 IoT sensor devices, including smoke density, carbon monoxide, and temperature sensors, are installed in each room and hallway. In this model, 19 zones are considered, which are connected to eight escape exits through egress routes (light green lines).

5.2. Preparation of fire knowledge base and test scenarios

To construct a rich fire knowledge base, numerous virtual fire scenarios were modeled and simulated considering various types of fires, as indicated in Table 5. As shown in Fig. 7(b), the initial locations of the virtual fires (depicted by red rectangles) were distributed in each room (216 points) and corridor (38 points). For each fire-initiation location, five fire classes were considered according to the NIST fire classification: ordinary combustible materials (Class A), flammable liquids (Class B), electrical equipment (Class C), combustible metals (Class D), and cooking oils (Class K). In addition, fire sizes of 0.5 m × 0.5 m, 1.0 m × 1.0 m, and 2.0 m × 2.0 m were applied to model the virtual fire scenarios. Accordingly, a total of 3810 (254 locations × five fire classes × three fire sizes) virtual fire scenarios were modeled and simulated to construct the fire knowledge base.

The experiments on the IDT-based FEMS were conducted with a number of test scenarios, assuming that fires occurred inside rooms and corridors, as shown by the yellow rectangle in Fig. 7(b). The detailed settings of the test scenarios are listed in Table 6. Considering safety and site constraints, we constructed simulation test scenarios instead of creating actual fire situations so that the sensor data acquired from these test scenario simulations were considered real sensor data.

Table 6
Consideration of fire class, fire size, and fire-initiation location for test scenarios.

Test scenario	Fire location	Fire class	Fire size (m)
A	Zone1: Inside room 1	A	1.0 × 1.0
B	Zone2: Inside room 12	C	0.5 × 0.5
C	Zone3: Corridor in front of room 54	A	1.0 × 1.0
D	Zone4: Inside room 166	K	2.0 × 2.0
E	Zone5: Inside room 21	D	0.5 × 0.5
F	Zone6: Corridor in front of room 68	A	1.0 × 1.0
G	Zone7: Inside room 171	K	1.0 × 1.0
H	Zone8: Corridor in front of room 72	A	2.0 × 2.0
I	Zone9: Inside room 136	K	2.0 × 2.0
J	Zone10: Corridor in front of room 143	A	1.0 × 1.0
K	Zone11: Corridor in front of room 35	A	1.0 × 1.0
L	Zone12: Inside room 86	B	0.5 × 0.5
M	Zone13: Corridor in front of room 188	A	2.0 × 2.0
N	Zone14: Corridor in front of room 190	A	1.0 × 1.0
O	Zone15: Corridor in front of room 97	A	1.0 × 1.0
P	Zone16: Corridor in front of room 103	A	1.0 × 1.0
Q	Zone17: Inside room 201	K	2.0 × 2.0
R	Zone18: Corridor in front of room 210	A	1.0 × 1.0
S	Zone19: Corridor in front of room 216	A	1.0 × 1.0

Table 7
Top five similar fire knowledge at $t = 30$.

Rank	Location	Fire class	Fire size (m)	DTW distance
1	Room 1	A	1.0 × 1.0	5.99
2	Room 1	B	1.0 × 1.0	6.10
3	Room 1	K	1.0 × 1.0	6.14
4	Room 1	D	1.0 × 1.0	6.21
5	Room 1	C	1.0 × 1.0	6.26

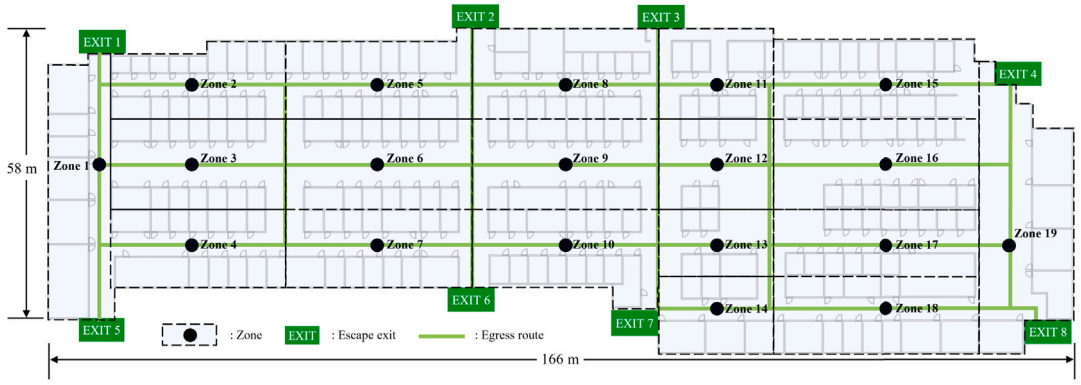
6. Results and analysis

6.1. Analysis of IDT-based FEMS on test scenarios

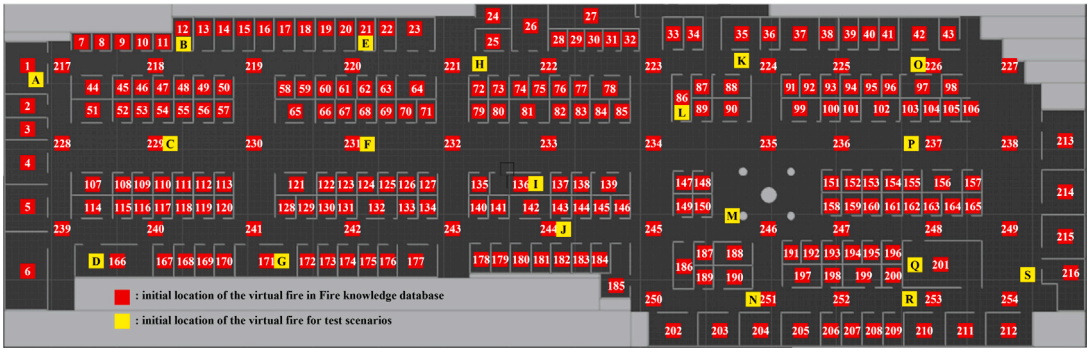
In Test Scenario A, it was assumed that a fire of Class A and size 1.0 × 1.0 initiated in Room 1, and 30 s passed since it was detected. From the recognition state, the IDT sequentially invoked the knowledge discovery, and building state similarity measure FEs to discover fire knowledge similar to the current building state. This discovery process took approximately 1 s on a Core i-9 3.5 GHz, thus satisfying the need to perform in near real time during a fire emergency. Table 7 lists the top five similar fire knowledge derived through online DTW-based similarity measurements. These represented the fire scenarios that occurred in Room 1. For the smallest DTW distance of 5.99, the most similar fire knowledge was recognized as a fire of Class A and size 1.0 × 1.0, which was identical to Test Scenario A.

In the response state, the ARET for each route was first calculated by comparing the untenability criteria with the recognized fire hazard factors using Table 4 and Eq. (5). Fig. 8 shows the untenable time of the area and some examples of the calculated ARET in Zones 1 and 19. As the fire spread, starting from the top-left corner, the untenable time showed low values near Room 1 and increased in the bottom-right direction. Accordingly, the calculated ARETs were less than 24 s in Zone 1, whereas they were more than 1000 s in Zone 19. For egress routes between Zone 1 and Exits 1, 2, and 5, the ARETs were 10 s, 20 s, and 24 s, respectively, indicating that the egress routes will become untenable in that order.

From the calculated ARETs for all egress routes, the RRET, SM, and safest routes were calculated using Eqs. (6), (11) and (12), respectively.



(a) Plain building layout and zone configuration.



(b) Initial location of the fire.

Fig. 7. Experimental settings for the case study.

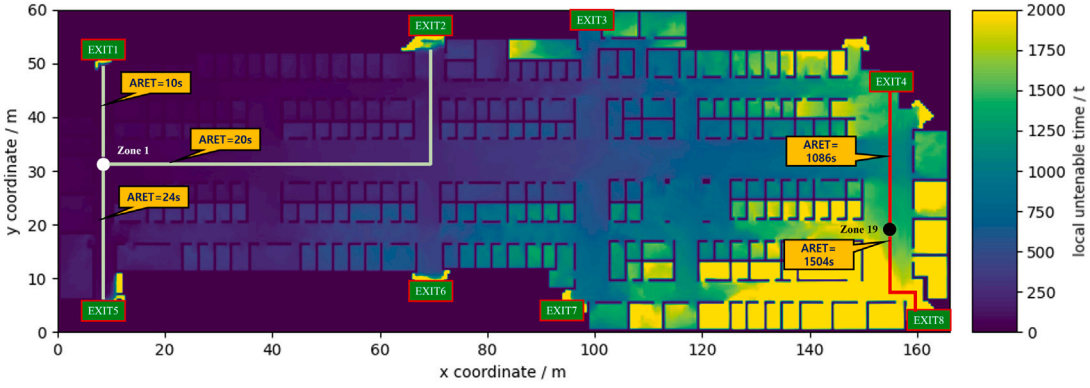


Fig. 8. Calculated ARET in test scenario A.

Table 8 lists the distances, ARET, RRET, and SM for egress routes to all exits in Zones 1–7, which are relatively more dangerous than the other zones. The red, blue, and purple colors of each cell represent the shortest route, safest route, and route that overlaps the shortest and safest routes, respectively. For Zones 1–3, which were close to the fire source, the SMs of all egress routes had negative values, indicating that people in those zones may be under threat by the fire while escaping. Therefore, the people in these zones must evacuate immediately or be rescued by fire responders. In such cases, the safest routes derived through the minimum SM can support safe evacuation and rescue, even if the distances are longer than the shortest route. Taking Zone 1 as an example, the safest route was tenable for 14 s longer than the shortest route (ARET > 14 s), and it took 12.5 s less to evacuate (RRET < 12.5 s) because of the relatively low impact of the fire on the movement speed. In Zone 4, evacuees can escape safely if they use the safest route rather than the shortest. As described in Table 8, the shortest route reached an untenable condition before the completion of escape (ARET = 42.0 s < RRET = 52.5 s), whereas the safest route allowed a marginal time of 90.8 s (ARET = 165.0 s > RRET = 74.2 s). Accordingly, when using the safest route, evacuees in Zone 4 can take pre-movement action, which is allowed up to 90.8 s. For Zones 5–7, which were far from the fire source, both the safest and shortest routes had positive SMs. Therefore, evacuees in these zones can escape safely and rapidly regardless of whether they choose the shortest or safest route.

The safety margin calculated for each zone can help firefighters to establish appropriate rescue strategies. Fig. 9 shows the maximum safety margin for each zone and the rescue priority derived based on the safety margin. The red-labeled zones on the left side of Fig. 9 have a negative safety margin, indicating that injury can occur regardless of the safest route used in these zones. Therefore, the occupants located in the red-labeled zones require immediate rescue. After the rescue of the most dangerous area is complete, rescue efforts prioritize the zones

Table 8
ARET, RRET, and SM of Zones 1 to 7.

Zone	Metrics	Exit 1	Exit 2	Exit 3	Exit 4	Exit 5	Exit 6	Exit 7	Exit 8
1	Dist	18.0	80.0	118.0	160.0	24.0	80.0	114.0	170.0
	ARET	10.0	20.0	20.0	20.0	24.0	20.0	20.0	20.0
	RRET	68.7	97.7	128.9	163.9	56.2	97.7	126.0	172.7
	SM	−58.7	−77.7	−108.9	−143.9	−32.2	−77.7	−106.0	−152.7
2	Dist	20.0	52.0	90.0	132.0	52.0	72.0	112.0	168.0
	ARET	10.0	31.0	31.0	31.0	10.0	31.0	31.0	31.0
	RRET	74.8	74.2	105.4	140.4	120.0	95.8	124.2	170.8
	SM	−64.8	−43.2	−74.4	−109.4	−110.0	−64.8	−93.2	−139.8
3	Dist	33.0	65.0	103.0	145.0	39.0	65.0	99.0	155.0
	ARET	10.0	60.0	60.0	60.0	20.0	60.0	60.0	60.0
	RRET	81.2	85.0	116.2	151.2	68.8	85.0	113.3	160.0
	SM	−71.2	−25.0	−56.2	−91.2	−48.8	−25.0	−53.3	−100.0
4	Dist	46.0	78.0	116.0	158.0	26.0	52.0	86.0	142.0
	ARET	10	132.0	165.0	165.0	42.0	165.0	165.0	165.0
	RRET	97.3	95.8	127.1	162.1	52.5	74.2	102.5	149.2
	SM	−87.3	36.2	37.9	2.9	−10.5	90.8	62.5	15.8
5	Dist	50.0	22.0	59.0	101.0	83.0	48.0	82.0	138.0
	ARET	10.0	78.0	78.0	78.0	42.0	78.0	78.0	78.0
	RRET	100.2	48.8	80.8	115.0	99.6	70.4	98.8	145.4
	SM	−90.2	29.2	−2.0	−37.0	−57.6	7.6	−20.8	−67.4
6	Dist	63.0	35.0	72.0	114.0	70.0	35.0	69.0	125.0
	ARET	10.0	132.0	179.0	179.0	42.0	179.0	179.0	179.0
	RRET	111.1	59.6	90.8	125.8	88.8	59.6	87.9	134.6
	SM	−101.1	72.4	88.2	53.2	−46.8	119.4	91.1	44.4
7	Dist	76.0	48.0	85.0	127.0	57.0	22.0	56.0	112.0
	ARET	10.0	132.0	362.0	362.0	42.0	362.0	362.0	362.0
	RRET	121.9	70.4	101.7	136.7	77.9	48.8	77.1	123.8
	SM	−111.9	61.6	260.3	225.3	−35.9	313.2	284.9	238.2

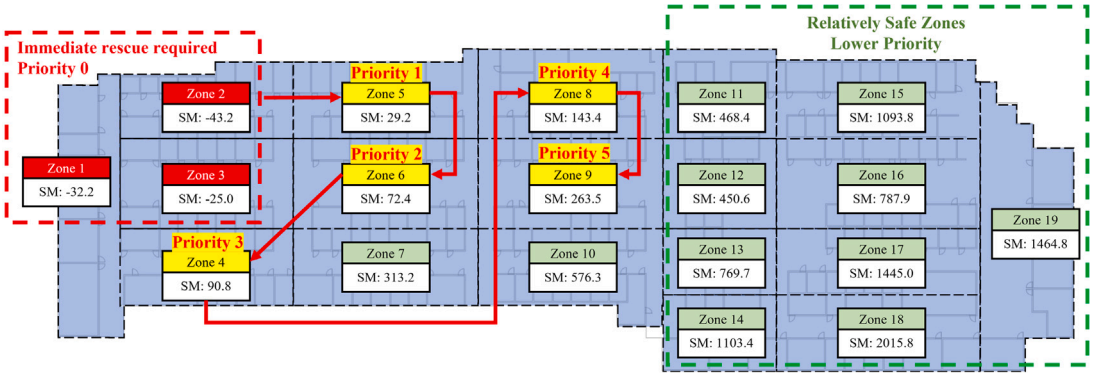


Fig. 9. Rescue strategy based on safety margin of the zones.

in ascending order of their safety margins. For example, as indicated by the yellow-labeled zones, rescue is first undertaken for Zone 5, which has a low safety margin of 29.2, followed by Zones 6, 4, 8, and 9, which have safety margins of 72.4, 90.8, 143.4, and 263.5, respectively. The safety margin represents the minimum possible time to escape from the building without injury; therefore, considering that time when determining rescue priorities can reduce the number of injuries.

To measure the impact of fires of different locations, classes, and sizes, IDT-based potential fire risk assessments were conducted for 19 test fire scenarios. Details of location, class, and size of fire in each test scenario are shown in Fig. 7(b) and Table 6. As a result of the similarity comparison, fire knowledge with the same fire class, fire size, and closest fire-initiation location was discovered in the corresponding test scenario, as depicted in the second column of Table 9 (labeled by their location-class-size). The lowest safety margins of the top-1 dangerous zone were observed in test scenarios C and F, where a fire was initiated in the corridor. This was due to the immediate blockage of escape routes, as they overlapped with the fire-spread paths in these scenarios, highlighting the need for firefighters to prioritize rescue efforts in these zones. When considering the fire size, the largest gap in safety margins among the top-1–3 dangerous zones was recorded in test scenarios B, E, and L, which had a small fire size of 0.5 × 0.5. Conversely, in Test Scenario M, in which a large fire occurred in the corridor, the smallest gap in the safety margin was observed. This indicates that fire size substantially affects the rate of spread, and smaller safety margin gaps necessitate a more urgent response in terms of rescue and evacuation strategies.

To evaluate the precision of the fire knowledge-based risk assessment, the safety margins calculated from the discovered fire knowledge and test scenarios were compared, as listed in Table 10. When there was fire knowledge closely resembling the fire-initiation location of the test scenario in the fire knowledge base, the errors in the derived safety margin were found to be less than 5.6 s. However, as the difference in the fire-initiation location increased, the error in the derived safety margin increased, reaching a limit of 33.1 s. These results imply that the more significant the differences in fire characteristics between the actual and virtual fires in the fire knowledge base, the more errors there may be in the derived potential risk assessments. Therefore, in such cases, expanding the fire knowledge base by thoroughly designing potential fire scenarios is necessary.

Table 9
Results of the potential fire risk assessment.

Test scenario	Discovered knowledge	Top-1 dangerous		Top-2 dangerous		Top-3 dangerous	
		Zone	SM	Zone	SM	Zone	SM
A (room)	1-A-1.0 × 1.0	2	−43.2	1	−32.2	3	−25.0
B (room)	12-C-0.5 × 0.5	2	−30.9	5	33.2	3	161.7
C (Corridor)	229-A-1.0 × 1.0	3	−96.6	1	−35.2	6	−5.6
D (room)	166-K-2.0 × 2.0	4	−77.6	3	−24.9	1	−12.4
E (room)	21-D-0.5 × 0.5	5	−43.0	2	59.9	8	107.4
F (Corridor)	231-A-1.0 × 1.0	6	−93.1	9	−13.5	3	−6.9
G (room)	171-K-1.0 × 1.0	7	−29.9	4	−12.5	6	6.4
H (Corridor)	221-A-2.0 × 2.0	8	−64.7	5	−59.5	6	−55.8
I (room)	136-K-2.0 × 2.0	9	−71.8	6	−45.3	12	−37.3
J (Corridor)	244-A-1.0 × 1.0	10	−65.5	7	−35.9	13	−23.8
K (Corridor)	224-A-1.0 × 1.0	11	−48.7	15	−23.3	12	−11.5
L (room)	86-B-0.5 × 0.5	11	−40.7	12	−15.6	8	91.7
M (Corridor)	247-A-2.0 × 2.0	17	−64.2	13	−60.8	12	−50.3
N (Corridor)	252-A-1.0 × 1.0	14	−47.6	13	−30.0	18	−22.7
O (Corridor)	226-A-1.0 × 1.0	15	−65.7	11	−3.6	19	23.8
P (Corridor)	237-A-1.0 × 1.0	16	−81.8	12	−5.5	19	6.8
Q (room)	201-K-2.0 × 2.0	17	−88.2	18	−48.4	19	−29.1
R (Corridor)	254-A-1.0 × 1.0	18	−58.6	17	−33.9	19	−18.1
S (Corridor)	255-A-1.0 × 1.0	19	−50.6	18	−46.8	17	−8.7

Table 10
Errors between safety margins calculated from the discovered fire knowledge and test scenario.

Scenario	A	B	C	D	E	F	G	H	I	J
Error (s)	4.0	5.6	0.8	8.8	2.5	1.2	2.6	16.3	2.4	1.6
Scenario	K	L	M	N	O	P	Q	R	S	
Error (s)	11.0	4.0	33.1	0.2	4	9.5	9.4	8.2	2.9	

Table 11
Comparison of processing times for fire simulation, original DTW, online DTW, and online DTW with parallel processing under the following conditions: the length of real sensor data $N = 10$, length of virtual sensor data $M = 3600$, and number of fire knowledge $k = 3810$.

Method	Processing time
Fire simulation	>8 h
Original DTW	781.71 s
Online DTW	37.68 s
Online DTW (parallel)	0.88 s

6.2. Effect of online DTW-based fire knowledge discovery

The proposed IDT-based FEMS utilizes a knowledge accumulation and discovery approach to leverage computationally intensive fire simulations in emergencies. To evaluate the effectiveness of Online DTW-based fire knowledge discovery, we compared the processing times of the fire simulation and similarity measurement algorithms, as shown in Table 11. This evaluation was conducted on a Core i-9 3.5 GHz computing system, and similarity measurement algorithms were performed under the following conditions: $k = 3810$ fire knowledge, length of real sensor data $N = 10$, and length of virtual sensor data $M = 3600$.

As shown in Table 11, the processing time of the fire simulation utilizing the FDS exceeded 8 h, indicating that conducting such simulations during a fire emergency is impractical. Although the original DTW required a relatively low processing time (781.71 s) in the similarity measurement, which is not trivial for application in time-critical situations where fire recognition must be performed within a short period of time after a fire is detected. However, the proposed online DTW significantly reduced the processing time to 37.68 s by calculating the DTW cost matrix in a cumulative manner, and the processing time could be further minimized to 0.88 s by introducing parallel processing.

During a fire, the length N of the real sensor data progressively increased, leading to a notable expansion in the processing time disparity between the original and online DTW, as illustrated in Fig. 10. Because the time complexity of the original DTW was $O(kNM)$, the processing time increased exponentially, requiring a significant time of 12 493.85 s at $N = 50$. In contrast, online DTW showed a constant processing time of 37.77 s on average when calculating the cost matrix by accumulating a $1 \times M$ partial cost matrix at each time instead of calculating the entire cost matrix of $N \times M$. Accordingly, online DTW can support real-time fire recognition with a consistent processing time that is not affected by the size of N.

6.3. Impact of zone-based risk assessment

To evaluate the impact of zone-based risk assessment, this study performed an evacuation simulation using Pathfinder [66]. The default settings of Pathfinder were used as parameters for all personal characteristics, and the change in movement speed due to fire in Eq. (7) was applied to the entire space. Fig. 11 illustrates a comparison of the evacuation time and injuries between escaping via the shortest and safest routes with varying occupant capacities (500, 1000, 2000, and 3000) in Test Scenario A.

The evacuation through the shortest routes required 94.3, 136.3, 227.0, and 323.5 s, whereas evacuation through the safest routes required shorter times of 85.8, 109.5, 175.0, and 258.5 s, respectively. Accordingly, the selection of the safest route could save up to 65 s when there were 3000 occupants in the building. This effect was mainly relevant to the occupants in Zones 1–3. In evacuation via the shortest route, occupants encountering the fire-spread path at Exit 1 experienced significant delays owing to the impact of the fire, prolonging the evacuation process. Conversely, in evacuation via the safest route, occupants quickly completed the evacuation by avoiding the fire-spread path as much as possible. Evacuation via the shortest route resulted in 73, 158, 366, and 708 injuries, whereas evacuation via the safest route resulted in 32, 73, 146, and 302 injuries, respectively. Utilizing the safest routes rather than the shortest routes reduced the number of injuries by up to 406. This is because the shortest route becomes untenable in Zones 1, 2, and 3 during evacuation, whereas the safest routes with a high ARET are relatively tenable for a long time.

7. Discussion, limitation, and future work

Regarding the RQ1 on providing appropriate fire situation information to stakeholders in large complex buildings, this study designed and developed an IDT-based FEMS. The system is based on an IDT, which has attributes, behaviors, and interfaces, and adopts a cloud–edge computing structure to manage the vast data generated in the building [51,52]. Through a case study of the Hoehyeon underground shopping mall, we examined in detail how situational information is

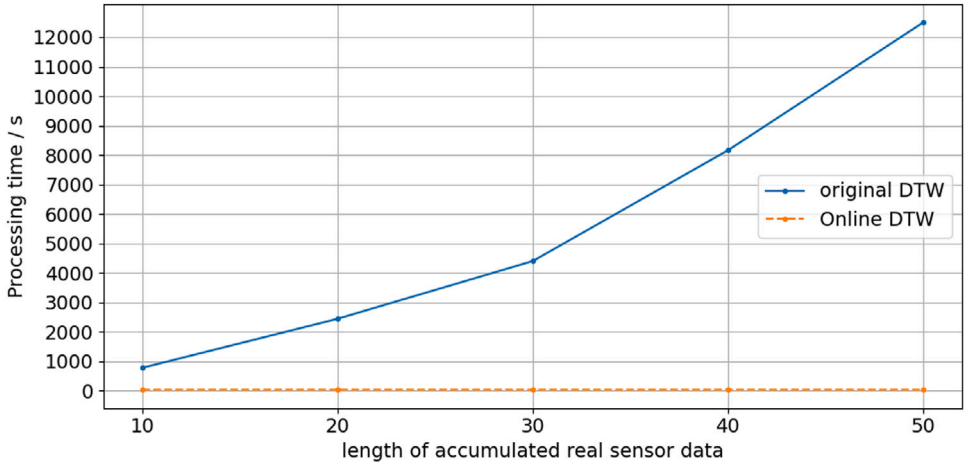


Fig. 10. Comparison of processing times for original DTW and online DTW with increasing length of real sensor data N.

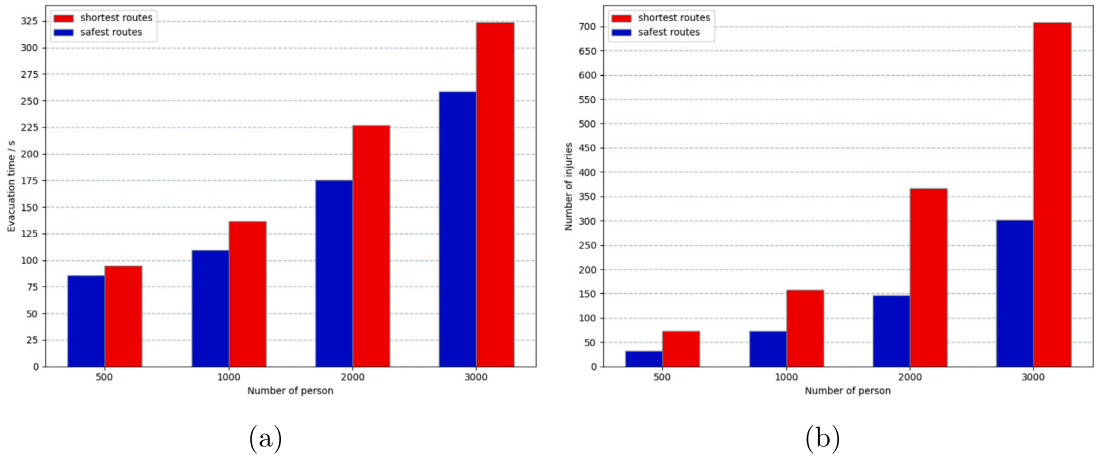


Fig. 11. Comparison of (a) the total evacuation time and (b) the total number of injuries between the shortest route-based and the safest route-based evacuation.

processed within the system and obtained (Section 6.1). Additionally, the development outcomes of applications linked to the system, presented in the appendix, address stakeholder concerns regarding the utilization of situational information.

Trends in fire and smoke spread provide crucial information for fire responses. However, owing to the long processing time of CFD simulations, this information is mainly used only in post-incident analyses [13]. To overcome this issue, we constructed a knowledge base to accumulate various fire scenarios through simulations and introduced a strategy to extract and use the scenario that was most similar to the ongoing fire situation. In this process, we proposed fire knowledge discovery algorithms that can find a similar scenario within 0.88 s. This overcomes the simulation execution speed issue, which requires over 8 h, enabling real-time utilization (Section 6.2). This answers the RQ2.

During a fire, safe areas at the current time can become hazardous as the fire spreads. Large complex buildings take longer to evacuate and rescue from than regular buildings. Therefore, potential risks must be considered to ensure safe evacuation and prompt rescue. Based on the guidelines in [61–63], we proposed a zone-based risk assessment algorithm that considers three fire-related risk factors. This algorithm calculates the safety margin time for each zone, enabling fire stakeholders to respond safely by considering the potential risks. Evacuation considering potential hazards resulted in a promising outcome, reducing the evacuation time by 65 s and decreasing the number of injuries by 406 compared to the shortest-path-based evacuation (Section 6.3). This answered RQ3.

Despite the promising results of the IDT-based FEMS, this study has three major limitations that must be addressed in future studies.

First, as the number of buildings managed by the proposed system increases and the number of sensors within the buildings increases, there is a potential burden on computing and network resources that could lead to system malfunctions [67]. We believe that the issue of computing resources can be mitigated by prioritizing critical tasks using container-based orchestrators [68], which are incorporated into our system design for efficient resource deployment and management. Running an IDT at the edge can reduce resource strain by saving network bandwidth; however, if urgent data transmission is not guaranteed, malfunctions may still occur. Applying priority-based networks [69] or time-sensitive networking technologies [70] to the network infrastructure of the system is a potential solution.

Second, the accuracy of the fire and smoke spread trends derived by the system depends on the knowledge accumulated in the knowledge base. An underdeveloped knowledge base can lead to failure of the fire response. We believe that future studies can construct a sufficiently developed knowledge base to handle various fire scenarios using the following techniques: (1) Setting a DTW distance threshold to alert experts regarding insufficient fire scenarios, prompting the generation of relevant scenarios, and (2) automatically generating fire scenarios using generative AI technologies [71]. The proposed system allows for large-scale knowledge base construction in the cloud and facilitates the replacement and updating of IDT FEs, making it easy to adopt these strategies and technologies.

Third, the fire risk assessment in this study has room for improvement as it was conducted in limited situations. The study considered only a limited number of fire risk factors during the risk assessment and assumed an average speed for human movement when calculating the RRET. This approach does not account for obstacles and blocked paths within buildings or individual physical characteristics such as age and disability. These limitations can be addressed by updating the fire risk assessment FE in the proposed system. For instance, in the former case, we can reflect the changes in building conditions perceived through spatial monitoring [72] in the ARET calculation (Eq. (5)) or derive the impact of state changes on the evacuation speed and incorporate it into the RRET calculation (Eq. (7)). In the latter case, we can optimize the evacuation and rescue plans by providing personalized risk assessment information based on detecting and tracking evacuees [23] within the building, considering individual differences rather than the average speed.

8. Conclusion

This paper presented an IDT-based FEMS for fire emergency management in large complex buildings. The IDT-based FEMS seamlessly integrates the cloud, edge, and device layers to ensure immediate access to fire information during emergencies. The IDT of the edge layer, built with an object-oriented design, efficiently gathers and processes data on the physical state of a building measured from the device layer. This process adheres to behavioral logic and encompasses detection, recognition, response, and treatment. These are executed using FEs that provide the necessary fire information for each phase of emergency management. For real-time fire recognition, the IDT-based FEMS uses a strategic approach for knowledge accumulation and discovery. In this approach, computationally intensive fire modeling and simulations can be executed in the cloud layer of the FEMS. Simultaneously, the edge-based IDT discerns similarities in the simulated fire knowledge relevant to the current building status. The similarity measurement between the real and simulated virtual sensor data employs an efficient online DTW algorithm. The IDT utilizes discovered fire knowledge to evaluate fire risks in various zones within a building by introducing metrics such as ARET, RRET, and SM between specific zones and exits. This enables the IDT to provide information on potential risks by zone, along with fire and smoke spread trends. We expect that the proposed IDT-based FEMS will significantly enhance building fire emergency management by providing crucial fire information during emergencies.

CRediT authorship contribution statement

Young-Jin Kim: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Validation, Visualization, Conceptualization. **Hanjin Kim:** Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Validation, Visualization. **Beomsu Ha:** Visualization, Validation, Data curation. **Won-Tae Kim:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. FEMS applications

Fig. A.12 shows the FEMS application provided by the IDT-based FEMS. To construct a fire knowledge base, fire experts model potential fire scenarios using Blender, a fire-scenario design application, as shown in Fig. A.12(a). Blender is an open-source three-dimensional modeling software that supports BIM model import in IFC format, FDS modeling, and FDS file export by applying the BlenderBIM and BlenderFDS plugins. When the fire scenario modeling is completed, the fire experts send the modeled FDS extension file to the Building Fire Emergency Manager through FDS Upload Web, as shown in the bottom-right corner of Fig. A.12(a). The Fire Scenario M&S FEs then run the FDS software using the FDS file and store the simulation results in the fire knowledge base.

Existing building fire-monitoring applications primarily provide visualized information by overlaying sensor data collected in real time on a virtual building. However, this only shows the current state of fire spread, making it difficult to preemptively understand the complex fire-spread patterns in complex buildings. Fig. A.12(b) shows an FEMS application that can be used in the fire recognition and response stages, visualizing the two-dimensional fire hazard factor information of the most similar fire knowledge through the web. In a fire situation, evacuees and rescuers can use the scrollbar at the bottom of the image to understand future fire spread trends in advance, allowing them to derive an optimal response strategy.

Fig. A.12(c) shows a structure support FEMS application utilizing IDT-based risk assessment results. Based on the risk assessment, the application displays the risk level of each area to help firefighters establish an efficient rescue strategy during the response and treatment stages. The risk level of each zone, derived through each area's safety margin, is categorized into three levels, "E1", "E2", and "E3", and is marked in the zones with red, yellow, and green labels, respectively. Firefighters can use this to prioritize areas with higher risk and proceed with rescue operations accordingly.

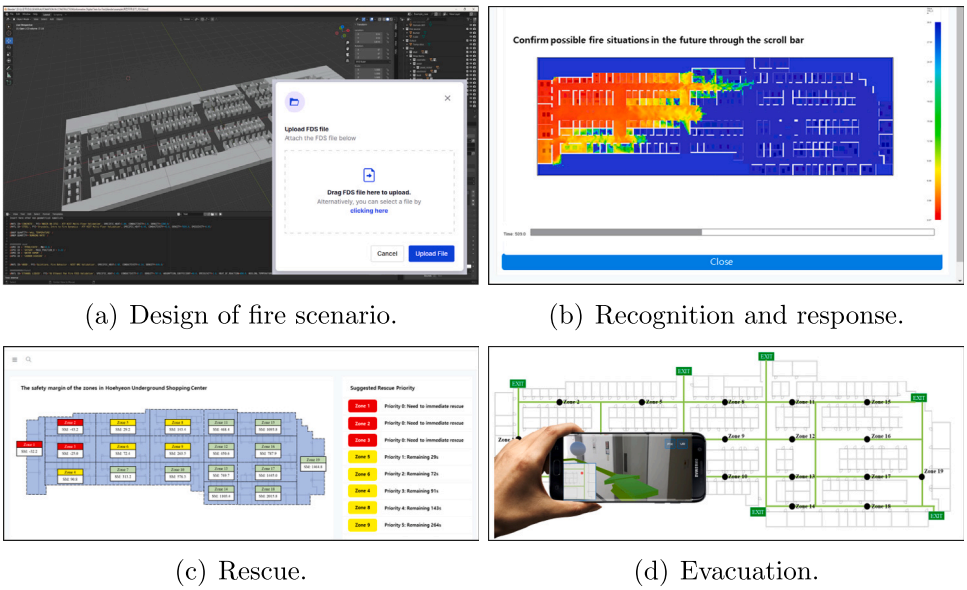
Fig. A.12(d) shows an FEMS application for evacuees that uses IDT-based risk assessment results. Similarly, the risk level information of each zone, derived from each area's safety margin, is used to calculate safe evacuation routes for evacuees. The AR application running on the smartphone displays the current location of the evacuee and the calculated safe route.

Appendix B. Fire prediction based on fire knowledge generated from the FDS

In this study, we predicted the expansion and migration of fire products based on fire knowledge generated by FDS. To confirm the validity of fire knowledge-based fire prediction, this Appendix introduces a real fire experiment conducted at the National Institute of Standards and Technology (NIST).

Fig. B.13 shows the building layout of a manufactured home, which was a test site constructed as part of NIST's Home Smoke Alarm Project. A temperature sensor (depicted as a black circle) was installed in Bedroom #1, and two real fire experiments were conducted with the fire-initiation point in the upper right corner of the living room and the upper side in Bedroom #1.

To predict fire behavior, we generated fire knowledge with fire-initiation points uniformly distributed across the building area using the FDS, which are depicted as triangles in Fig. B.13.



(a) Design of fire scenario. (b) Recognition and response.

(c) Rescue. (d) Evacuation.

Fig. A.12. Applications provided by the IDT-based FEMS.

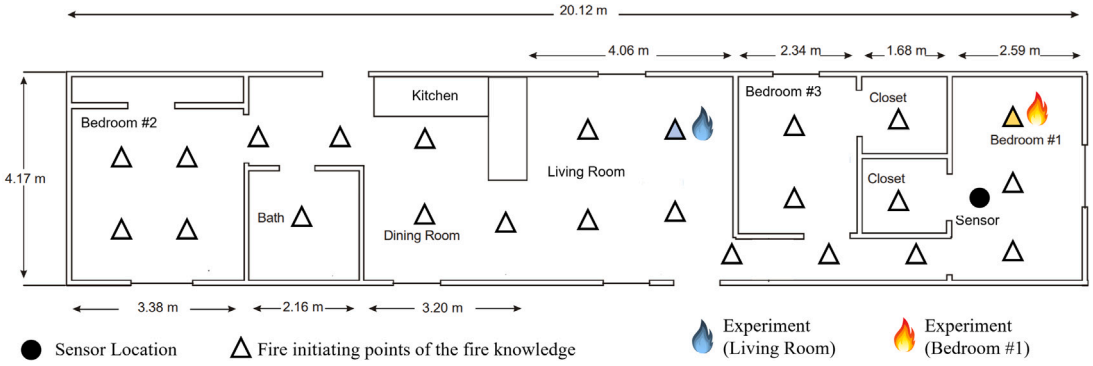


Fig. B.13. Building layout of the manufactured home in NIST's home smoke alarm project.

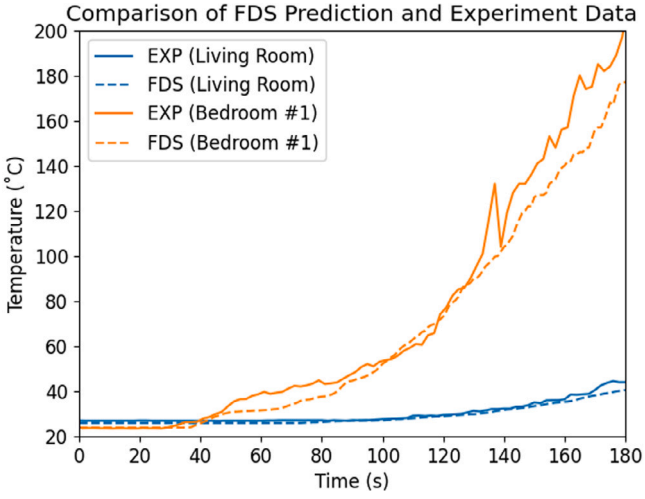


Fig. B.14. Building layout of the manufactured home in NIST's home smoke alarm project.

Comparing the fire status similarities in this experimental setting, the fire knowledge closest to the actual fire location was identified as the most similar, as depicted by the blue and orange filled triangles in Fig. B.13. A comparison of temperature data between most similar fire knowledge and experimental data is shown in Fig. B.14. The solid lines refer to real fire experiments, the dotted lines refer to fire knowledge, and the blue and orange colors represent the fire scenarios initiated in the living room and Bedroom #1, respectively.

In the temperature data presented in Fig. B.14, both the experimental data and fire knowledge of the Bedroom #1 fire started to increase simultaneously at 40 s and reached the untenable threshold of 100 °C at approximately 134 s, with a small difference of 3 s. In the case of the living room fire, both the experimental data and fire knowledge started to gradually increase from 100 s, with almost identical trends. These experiments inspired us to use FDS-based fire knowledge to predict building fires.

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